

Engineering Standard

14 May 2023

SAES-X-600

Cathodic Protection of External Surfaces of Plant Facilities

Document Responsibility: Cathodic Protection Standards Committee

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Summary of Changes

Paragraph Number		Change Type (Addition, Modification, Deletion, New)	Technical Change(s)
Previous Revision 01 February 2021	Current Revision 14 May 2023		
5.1.2.4 5.3.5.3	5.1.2.4 5.3.5.3	Addition	Remote Monitoring Units (RMU)
5.2.1 5.3.1	5.2.1 5.3.1	Addition	Continuous linear anode included
5.2	5.2	Addition	Design current density for copper in aerated environment added.
Table-4	Table-4	Modification	MMO Anodes requirements referred to 17-SAMSS-007
5.2.7.7 5.4.3.1.2	5.2.7.7 5.4.3.1.2	Modification	Junction boxes requirement referred to 17-SAMSS-008
5.2.13.6	5.2.13.6	Addition	Solid state polarization cells included
5.3.4.5	5.3.4.5	Modification	For multiple short sections with combined length of 500m and more, ICCP is the preferred method

1 Scope

- 1.1** This standard prescribes the minimum mandatory requirements governing the design and installation of cathodic protection systems for the external surfaces of buried and submerged structures for onshore plant facilities. This standard does not address buried pipelines entering or exiting the plant before the launcher/receiver area.
- 1.2** Plant facilities that require cathodic protection include but are not limited to the following structures:
- a. Buried pressurized piping including short sections at road crossings, dike crossings, or at connections to existing underground piping.
 - b. Buried non-pressurized piping containing a product that might pose a safety or environmental concern if leakage was to occur, i.e., gravity drain lines with H₂S.
 - c. Only if economically viable and when stray current corrosion could be a concern, non-pressurized buried hydrocarbon piping shall be electrically bonded to the CP systems, but does not need to comply with any protection criteria.
 - d. Vent pipes and non-pressurized instrument air pipes do not require cathodic protection.
 - e. Pipeline casings for pressurized piping at road or dike crossings.
 - f. The bottom or soil-side of above ground storage tanks.
 - g. Buried metallic storage tanks.
 - h. Critical metal piles, onshore side of sheet piling and associated anchors. Non-critical metal piles shall be coated for the entire length with APCS-113A as per SAES-H-002 and do not require cathodic protection.
 - i. Buried valves and fittings connected to non-metallic underground piping.
 - j. Water wells inside the plant facilities shall be protected as detailed in SAES-X-700.
 - k. Catch basins and metal fittings and appurtenances associated with catch basins are outside the scope of this standard.
- 1.3** This standard may not be attached to nor made part of purchase orders.

2 Conflicts and Deviations

Any conflicts between this document and other applicable Mandatory Saudi Aramco Engineering Requirements (MSAERs) shall be addressed to the EK&RD Manager.

Any deviation from the requirements herein shall follow internal company procedure SAEP-302.

3 References

All referenced specifications, standards, codes, drawings, and similar material are considered part of this Engineering Standard to the extent specified, applying the latest version, unless otherwise stated.

3.1 Saudi Aramco References

Saudi Aramco Engineering Procedures

SAEP-302	Waiver of a Mandatory Saudi Aramco Engineering Requirement
SAEP-303	Engineering Reviews of Project Proposal and Detail Design Documentation

Saudi Aramco Materials System Specification

17-SAMSS-004	Tap Adjustable Rectifiers for Cathodic Protection
17-SAMSS-005	Cathodic Protection Phase Controlled Rectifiers
17-SAMSS-006	Galvanic Anodes for Cathodic Protection
17-SAMSS-007	Impressed Current Anodes for Cathodic Protection
17-SAMSS-008	Junction Boxes for Cathodic Protection

17-SAMSS-017	Impressed Current Cathodic Protection Cables
17-SAMSS-018	Remote Monitoring System (RMS) for Cathodic Protection Applications
32-SAMSS-005	Manufacture of Atmospheric Tanks

Saudi Aramco Engineering Standards

SAES-H-002	Internal and External Coatings for Steel Pipelines and Piping
SAES-X-400	Cathodic Protection of Buried Pipelines
SAES-X-700	Cathodic Protection of Onshore Well Casings
SAES-D-100	Design Criteria of Atmospheric and Low-Pressure Tanks

Saudi Aramco Best Practices

SABP-X-003	Cathodic Protection Installation Requirements
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Saudi Aramco Standard Drawings

AA-036276	Multi-purpose Junction Box Details
AA-036277	5 Terminal Bond Box Details
AB-036346	Surface Anode Bed Details - Horizontal and Vertical Anodes (Sheets 1&2)
AA-036349	3 Terminal Bond Box Details
AA-036350	2 Terminal Bond Box Details
AA-036352	Galvanic Anodes for Roads and Camel Pipeline Crossing
AA-036355	Tank Bottom Impressed Current Cathodic Protection
AB-036378	Cathodic Protection Onshore Rectifier Installation Details
AB-036381	Thermite Welding of Cables to Pipelines and Structures
AA-036385	Deep Anode Bed without Anode Support Pipe
AA-036629	Anode Details for Hydrants and Risers from RTR or Plastic Mains
AA-036637	Bond Box Details for Manholes
AD-036785	Symbols for Cathodic Protection
AA-036905	Details of Installation of New Bottoms and Cathodic Protection for Existing Welded and Riveted Tanks
AB-036907	Kilometer Marker & CP Test Stations for Buried Pipelines

3.2 Other References

National Fire Protection Association

NFPA 70	National Electric Code (NEC)
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The Association for Materials Protection and Performance (AMPP/NACE)

AMPP SP0169-2013	Control of External Corrosion on Underground or Submerged Metallic Piping Systems
AMPP SP0193 – 2016	External Cathodic Protection of On-Grade Carbon Steel Storage Tank Bottoms
AMPP SP0285 – 2021	Corrosion Control of Underground Storage Tank Systems by Cathodic Protection
AMPP SP0200 – 2014	Steel-Cased Pipeline Practices

- AMPP SP0286 – Electrical Isolation of Cathodically Protected Pipelines
2007
- AMPP TM0497 Measurement Techniques Related to Criteria for Cathodic Protection on
– 2012 Underground or Submerged Metallic Piping Systems

International Organization of Standardization

- ISO 15589-1- Petroleum and Natural Gas Industries - Cathodic Protection of Pipeline
2015 Transportation Systems - Part 1: On-land Pipelines

The International Electrotechnical Commission

- IEC 62305-4 Protection against Lightning — Part 4: Electrical and Electronic Systems
within Structures

4 Terminology

4.1 Acronyms

ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers (USA)
ASTM	American Society for Testing and Materials
AWS	American Welding Society
CP	Cathodic Protection
CSD	The Saudi Aramco Consulting Services Department
CSE	Saturated copper/copper sulfate reference electrode
GOSP	Gas and Oil Separation Plant
HDD	Horizontal directional drilling
ICCP	Impressed Current Cathodic Protection
MSAERs	Mandatory Saudi Aramco Engineering Requirements
NEC	National Electric Code
NEMA	National Electrical Manufacturers Association (USA)
OHPL	High-voltage overhead or buried power lines
RSA	Saudi Aramco Responsible Standardization Agency
SAES	Saudi Aramco Engineering Standard
SAMSS	Saudi Aramco Materials System Specification
SCC	Stress corrosion cracking
WIP	Water Injection Plant

4.2 Definitions

- AA:** The ANSI cooling class for a dry-type self-cooled transformer or reactor that is cooled by the natural circulation of air. Syn.: AN (IEC), See definition 3.1 in IEEE C57.12.

Anode:	An impressed current or a galvanic anode for cathodic protection applications.
Anode Cable:	A cable directly connected to an impressed current or galvanic anode.
ANV:	The ANSI cooling class for a dry-type non-ventilated self-cooled transformer, which is so constructed as to provide no intentional circulation of external air through the transformer, and that operates at zero-gauge pressure. Syn.: ANAN (IEC), See definition 3.15 in IEEE C57.12.
Armored Cable:	Armored cable for cathodic protection is a single core insulated cable manufactured with a double layer spiral wound insulated galvanized steel sheath to provide mechanical protection, and typically used for subsea applications.
Bond Cable:	A cable installed between two metallic structures to provide electrical continuity between the structures for the purpose of cathodic protection.
Buyer:	Saudi Aramco Purchasing Department Representative
Buyer's Representative:	The person or persons designated by the Purchasing Department to monitor/enforce the contract. Normally, this is the on-site inspector.
Calcined Petroleum Coke Breeze:	A carbonaceous backfill used as a conductive backfill media for impressed current anodes in soil.
Coated Casing:	The term "coated casing" as used in this engineering standard describes a well casing with an external non-conductive coating (typically Fusion Bond Epoxy or FBE). The coating must be applied to all sections of the casing in contact with soil or formation, from surface to the bottom of the casing or to a depth determined to facilitate external corrosion mitigation with cathodic protection through the relevant down hole corrosive aquifers. Casings that have been coated over the upper two or three joints of casing only are not "coated casings". Coating applied to well casings is not applied as a corrosion barrier. It is applied to reduce the total amount of CP required or to extend the influence of the applied CP.
CP Assessment Probe:	A CP assessment probe is a multi-electrode probe designed to enable measurement of the soil resistivity in addition to representative polarized and depolarized potentials for the pipeline or other buried or immersed metallic structure at the probe location.
CP Coupon:	A CP coupon is a single electrode coupon that has been designed to enable measurement of representative potentials, current densities, or corrosion rates on a pipeline or other buried or immersed metallic structure at the coupon location.
CP System Operating Circuit Resistance:	The total resistance seen at the output of a CP power supply or the total working resistance in a galvanic anode system.
CP System Rated Circuit Resistance:	The rated output voltage of a cathodic protection power supply divided by the rated output current. For photovoltaic power supplies, the rated output current is the design commissioning current.

Cross Country Pipeline:	A pipeline between; two plant areas, another cross country pipeline and a plant area, or between two cross country pipelines.
Deep Anode Bed:	Anode or anodes connected to a common CP power supply and installed in a vertical hole with a depth exceeding 15 m (50 ft).
Design Agency:	The organization responsible for the design of the pipeline and associated CP system. The Design Agency may be the Design Contractor, the Lump Sum Turn Key Contractor or an in-house design organization of Saudi Aramco.
Drain Point:	The location on the cathodically protected structure where the negative cable from the T/R or negative junction box is fastened to the structure.
ER Probe:	Monitoring probe that uses electrical resistance (ER) technique as an on-line method of monitoring the rate of corrosion and the extent of total metal loss for any metallic equipment or structure.
Flowline:	A pipeline connected to a well.
Foundry (for anodes):	An anode foundry is a facility that produces metal castings from either ferrous or non-ferrous metals or alloys.
Galvanic Anodes:	Anodes fabricated from materials such as aluminum, magnesium or zinc that are connected directly to the buried structure to provide cathodic protection current without the requirement for an external cathodic protection power supply. Galvanic anodes are also referred to as sacrificial anodes.
Hazardous Areas:	Those areas where fire or explosion hazards may exist due to flammable gases or vapors, flammable liquids, combustible dust, or ignitable fibers or filings (see NEC Article 500).
Impressed Current Anodes:	Anodes typically fabricated from High Silicon Cast Iron (HSCI) or Mixed Metal Oxide (MMO) that are connected through a DC power supply to the buried or immersed structure to provide cathodic protection current.
Instant-OFF Potential:	OFF potential measured with a short delay after interruption (switching off) of all sources of applied cathodic protection current with the aim of approaching the IR-free potential.
Isolating Spark Gap (ISG):	Component to discharge high voltages from lightning strikes and/or AC fault currents for isolating electrically conductive installation sections to prevent potential damage to isolation devices.
Lead Wire (for anodes):	A cable directly connected to an anode.
Manufacturer:	The Company that assembles the components for the finished product and provides the finished product either through a Vendor or directly to Saudi Aramco.
Mill (for anodes):	An anode mill is a facility that produces anodes such as MMO or Platinized Niobium (without lead wires) for cathodic protection applications (or any other anode type that does not involve casting).
Negative Cable:	A cable that is electrically connected (directly or indirectly) to the negative output terminal of a cathodic protection power supply or to a

	galvanic anode. This includes bond cables to a cathodically protected structure.
Non-incendive Equipment:	Equipment having electrical/electronic circuitry that is incapable, under normal operating conditions, of causing ignition of a specified flammable gas-air, vapor-air, or dust-air mixture due to arcing or thermal means. (See NEC Article 100).
Off-Plot:	Off-plot refers to any area outside of the plot limits.
ON Potential:	Potential measured while all sources of applied cathodic protection current (CP System) is continuously operating.
OFF Potential:	Potential measured after interruption (switching off) of all sources of applied cathodic protection current with the aim of approaching an IR-free potential.
ONAN:	The ANSI cooling class for a transformer or reactor having its core and coils immersed in mineral oil or synthetic insulating liquid with a fire point less than or equal to 300°C, the cooling being effected by the natural circulation of air over the cooling surface. (ONAN was previously termed OA). See definition 3.303 in IEEE C57.12.
On-Plot:	On-plot refers to any area inside the plot limits.
Perimeter Fence:	The fence which completely surrounds an area designated by Saudi Aramco for a distinct function.
Photovoltaic Module:	A number of solar cells wired and sealed into an environmentally protected assembly.
Pipeline:	The term “pipeline” is used generically in this standard and can be used to refer to any type of pipeline.
Plant Area:	A plant area is the area within the plot limits of a process or storage facility.
Plot Limit:	The plot limit is the boundary around a plant or process facility. The plot limit may be physical such as a fence, a wall, the edge of a road or pipe rack, chains and posts or a boundary indicated on an approved plot plan.
Positive Cable:	A cable that is electrically connected (directly or indirectly) to the positive output terminal of an impressed current cathodic protection power supply, including impressed current anode cables.
PMT:	Project Management Team used as a truncated version of Saudi Aramco Project Management Team or SAPMT.
Process Pipeline:	A pipeline typically associated with a plant process and typically above ground within a plant facility.
Production Pipeline:	A pipeline transporting oil, gas or water to or from a well. These include flow-lines, test-lines, water injection lines and trunk-lines.
Pyramid Anode:	An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and placed on the sea bed. The pyramid anode derives its name from the pyramid shaped concrete base.
Reference Electrode:	An industry standardized electrode used as a common reference potential for cathodic protection measurements. A copper/copper

	sulfate (Cu/CuSO ₄) reference electrode is typically used for soil applications. A silver/silver chloride (Ag/AgCl/0.6M Cl) reference electrode is typically used for aqueous applications.
Resistivity Meter (such as Megger Tester or equivalent):	A meter designed to measure ground resistivity, or can be connected to measure resistance in a format that excludes the resistance of the test wires.
Soil Transition Point:	The surface location where a pipeline enters or exits the soil, i.e., above grade to below grade transition, or below grade to above grade transition.
Stray Current:	Current flowing through the unintended circuits/paths.
Subject Matter Expert (SME):	For the purposes of this standard, the SME shall be the assigned Consulting Services Department cathodic protection specialist.
Surface Anode Bed	Anode or anodes connected to a common CP power supply, installed either vertically or horizontally at a depth of less than 15 m (50 ft).
Surge Diverters (SD):	Component to discharge high voltages from lightning strikes and/or AC fault currents for isolating electrically conductive installation sections to prevent potential damage to isolation devices.
Tension Spring Anode:	An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and suspended on a wire rope/tension spring assembly beneath an offshore platform.
Test Line:	A pipeline that is used for testing an individual well or group of wells:
Thermite Weld:	An exothermic process for use in making electrical connections between two pieces of copper or between copper and steel.
Transformer Rectifier (T/R):	A cathodic protection power supply that transforms and rectifies AC power into an adjustable output DC power for cathodic protection applications. Typically abbreviated or truncated as "T/R" or "Rectifier".
Transmission Pipeline:	A cross country pipeline transporting product between GOSPs WIPs or other process facilities.
Trunkline:	A pipeline designed to distribute or gather product from two or more wells, typically connecting flowlines or injection lines to the associated GOSP or WIP.
Uncoated Casing:	The term "uncoated casing" when used in this engineering standard describes a well casing that is either bare (no external coating), or may have an external coating applied to the shallow sections of casing to minimize corrosion in the landing base or surface soils.
Utility-line:	A pipeline designed to deliver an end use service product (typically water, gas or air).
Vendor:	A company that receives a purchase order to supply the finished product, material or equipment. The Vendor may also be the Manufacturer.

5 Design Requirements

5.1 General Design Requirements

5.1.1 Contractor/Designer Qualifications

5.1.1.1 Cathodic protection designs shall be carried out by design offices that are approved by Saudi Aramco. Such design offices must have qualified engineers with a minimum of five years verifiable cathodic protection design experience and a minimum industry qualification of NACE CP Level 4 and BS degree. NACE CP certification level, certification number, name of designer, and his signature/initial shall be provided in the design package.

Note:

Review, validation and approval of 90% detailed design package is responsibility of CSD and it is not delegated to anyone (including the approved designers).

5.1.1.2 Field measurements required for the design shall be collected by an Engineer or Technician with a minimum of five years verifiable cathodic protection field experience and industry certification level of NACE CP Level 2 or higher. Installation supervision of cathodic protection materials shall be done by Engineer or Technician with a minimum of five years verifiable cathodic protection field experience and industry certification level of NACE CP Level 2 or higher.

Note:

PMT may request assistance from CSD for the verification of the qualifications of the Design Contractor's engineer/technicians responsible for the CP field and engineering work.

5.1.2 Design Review

5.1.2.1 Proposed construction drawings and related cathodic protection design information in accordance with SAEP-303 shall be submitted to the Proponent cathodic protection organization and to Saudi Aramco's Consulting Services Department (CSD) for review and approval.

5.1.2.2 The design agency shall not issue drawings for construction until the design has been reviewed and approved in writing by CSD and the Proponent cathodic protection organization.

Exception:

Design package approval is not required for CP designs based specifically on a CP standard drawing, e.g., galvanic anode installations per Standard Drawing AA-036352, for road crossings on normally above grade pipelines with no dedicated CP systems.

5.1.2.3 The design package submitted for review shall contain at minimum:

- a. The scope of work
- b. Professionally drafted full size Index "X" CP drawings that:
 - Detail each CP item by description and stock number if applicable.
 - Detail the proposed location for each piece of CP equipment including but not limited to rectifiers, anodes, junction boxes, bond stations, test stations, soil access holes, bonds to pipelines, insulating flanges and unions.
 - Detail and clearly illustrate an organized numbering system (sequential order) for test points and soil access boxes that begins at the top left corner of the cathodic protection layout drawing.
 - Detail and specifically identify all cathodic protection cables including all anode, structure, bond, and rectifier cables. These details shall be shown on the One-Line Diagram for Impressed Current Cathodic Protection (ICCP) system.
 - Clearly identify the specific and individual cable routing and termination points within the respective test stations, junction boxes, bond boxes, and rectifiers.
 - Detail all cathodic protection equipment using the cathodic protection symbols as shown on Standard Drawing AD-036785 "Symbols for Cathodic Protection."
 - Detail all existing CP installations in the plant that could have an effect on the new CP system, and also provide information on spare current capacity of existing CP systems that will be utilized by the project.

- For plant piping CP, A0' sized plot plan drawing showing anode layout, anode gradient circles (anode voltage cones) and UG piping shall be provided with the design for review.
- Plot plan with hazardous area overlay drawing showing clearly all structures that are CP protected and all CP equipments and their locations shall be provided.
- c. All calculations and applicable field data are required to verify design compliance with the Saudi Aramco Cathodic Protection Engineering Standards. CSD's approval of the design calculations, using average soil resistivity values, does not relieve the designer from his responsibility of meeting the required potential criteria stated by this standard.

Notes:

1. *Cathodic protection design considerations begin at the Design Basis Scoping paper (DBSP) stage and shall be reasonably developed by the Project Proposal stage. The Project Proposal Design Package shall include all design considerations that can be developed without acquiring actual field measurement data, including spare system capacity within the plant, proposed anode type(s) and sizing of the cathodic protection power source(s).*
2. *The Detailed Design Package submitted to CSD for review, shall present a complete and comprehensive cathodic protection system design including all required field measurement data, calculations, a detailed description of the proposed cathodic protection equipment, and a set of detailed drawings that illustrate the proposed placement of all cathodic protection equipment and the associated protected structures.*
3. *It is recommended that the CP design office hold a preliminary meeting jointly with CSD and the Proponent CP Group to outline their design approach for the Project Proposal or Detailed Design CP package, before starting work on the design package. This will help to eliminate a large number of the design review comments resulting from inadequate field information or lack of in-depth existing CP system information, and speed up the review and approval processes.*

- d. The CP system shall be:
 - Galvanic, or
 - AC powered impressed current considering
 - Air cooled, or
 - Oil cooled T/R
 - Single phase, or
 - Three phase T/R.

5.1.2.4 New CP systems installed in a Plant area shall be provided with Remote Monitoring Units (RMU), with signal transmitters to the existing DCS or SCADA system. Remote monitoring equipment shall be provided for ICCP power supplies.

5.1.2.4.1 The remote monitoring equipment shall be constructed to interface with the existing remote monitoring system being used by the respective CP Proponent Department. Interfacing of the new & existing RMU is mandatory to ensure that the RMU data is reliably collected and utilized. To address this requirement, details of the communication system that will be used for the remote monitoring system and software that will be supplied for the CP Proponent shall be detailed. If the data is not transmitted to the Saudi Aramco data base, then the software shall be provided through the Project.

5.1.2.4.2 PMT shall determine the specifications for the existing remote monitoring system and shall be fully responsible for providing this information to the CP Design Contractor and ICCP power supply vendor.

5.1.2.4.3 In addition to the RMU at power supplies, at least remote monitoring unit shall be provided for the plant piping network in standalone area or when the total piping length is 500 m. This will result in multiple RMUs. For example, if the total piping length is 2000 m, then four RMUs will

- be required. Irrespective of the total piping length; each of the different plant areas (such as utility area and other plant areas,) must be provided with at least one RMU.
- 5.1.2.4.4** Remote Monitoring Systems shall comply with 17-SAMSS-018 and shall be constructed to interface with the existing Saudi Aramco Remote Monitoring Systems.

5.2 Technical Design Requirements

5.2.1 Design Philosophy

Unless otherwise specified in this standard, cathodic protection systems shall be designed using the "earth potential rise" method with distributed, continuous linear anode or with MMO grid type impressed current anodes.

Exception:

An alternative design philosophy using a combination of remote and distributed anode beds may be more cost effective in some situations. Proposals must be submitted with a complete and thorough economic comparison and shall be considered unacceptable unless pre-approved in writing by the proponent.

Note:

In the case of plant piping, the earth potential rise calculation is applicable as per Appendix I. For the other type of system such as linear anode and grid type under tank bottoms anode separation is the limiting factor as calculated within the permitted maximum as per applicable company drawings.

5.2.2 Protection Criteria

- 5.2.2.1** The potential criteria for the specific structures shall be achieved in the design by determining the number, placement, current loading, and length of the anodes required to achieve the necessary voltage gradient. The calculation method for this procedure shall use the "earth potential rise" equations detailed in Appendix I. To achieve the potential criteria for the specific structure, use the detailed values in the sections of this standard listed in Table 1.

- 5.2.2.2** Unless actual field data indicates otherwise "earth potential rise" calculations shall presume an initial structure potential of:

- 0.250 volt (Cu-CuSO₄) for structures electrically continuous with bare stainless steel or copper piping, bare ground cables (copper or tinned), or copper or stainless steel ground rods.
- 0.500 volt (Cu-CuSO₄) for structures that are not shorted to buried and bare copper or stainless steel structures, but may be electrically continuous with insulated ground cables, galvanized ground rods, or other galvanized or coated structures.

5.2.3 Design Current Density

Cathodic protection designs shall provide the minimum current densities detailed in Table 2.

5.2.4 Design Life

- 5.2.4.1** The minimum design life for galvanic anode systems shall be 15 years.

- 5.2.4.2** The minimum design life for impressed current systems shall be 25 years.

- 5.2.4.3** The design life for a galvanic system shall be calculated using parameters from Table 3 and the equation detailed below:

$$\left[\frac{\text{Total Weight of All Anodes(kgs)} \times \text{Utilization Factor}}{\{\text{Consumption Rate} \times \text{Calculated Anode output current for all anodes} *\}} \right] \geq \text{Design Life Years}$$

*For the life calculation of a galvanic anode bed system, use the lowest measured resistivity value for the planned installation depth of the anodes based on the finished grade levels at the site.

Notes:

1. *Design life stated above for galvanic anodes is not applicable for temporary CP systems if considered for the project.*
2. *The purpose of the above formula is to ensure that each anode must serve the required life taking into account its calculated current output, whilst the anodes will last longer if they operate less than the calculated current output. This has nothing to do with the cathodic protection current requirement.*

5.2.4.4 The design life for an impressed current distributed anode system shall be calculated using parameters detailed in Table 4 and the following equation:

$$\left[\frac{\text{Anode Weight (kgs)}}{(\text{Anode Consumption Rate} \times \text{Anode Current Capacity})} \right] \geq \text{Design Life Years}$$

The design life for an impressed current remote anode system shall be calculated using parameters detailed in Table 4 and the following equation:

$$\left[\frac{\text{Total Anode Weight (kgs)}}{(\text{Anode Consumption Rate} \times \text{Rectifier Rated Current Capacity})} \right] \geq \text{Design Life Years}$$

Use of other types of impressed current anode materials such as polymer anodes, for use in special applications may be permitted after thoroughly evaluating the anode performance. The impressed current anodes most commonly used by Saudi Aramco are as listed in Table 5.

5.2.5 Circuit Resistance

5.2.5.1 For a T/R, the CP system rated circuit resistance shall be defined as the T/R rated voltage, divided by the T/R rated current. Rated voltages and currents are as detailed on the manufacturer's data sheet/plate.

5.2.5.2 The CP system operating circuit resistance for an ICCP system shall be defined as the total effective resistance as seen by the output terminals of the respective ICCP power supply, and for calculation purposes shall include:

- a. Anode bed resistance to ground.
- b. Positive cable resistance from CP power source to anodes.
- c. Negative cable resistance from CP power source to structure.
- d. Resistance of the structure to remote earth.
- e. Effective resistance caused by +0.8 volts anode bed back emf (for HSCI in coke breeze) plus the structure back emf of -1.2 volts = 2.0 volts total between the anode with coke breeze backfill, and a coated buried piping.

$\text{Remf} = (0.8 \text{ v} + \text{back emf}) / I_{\text{rated}}$

$I_{\text{rated}} = \text{CP power supply "rated" current.}$

Exception:

A back emf of +1.0 volts (Cu/CuSO₄) shall be used for mixed metal oxide anodes in water tanks, subkha, or comparable high salinity wet applications without coke breeze. This does not include the back emf associated with the polarized potential of the structure.

Note:

When using Remf to calculate the maximum allowable anode bed resistance to provide 70% of Rated, the value of Remf must be multiplied by 0.7.

Example: $R_{\text{anodebed maximum}} = (0.7 \times (I_{\text{rated}} - \text{Remf})) - R_{\text{cables}} - R_{\text{structure}}$

- 5.2.5.3** ICCP system designs shall take into consideration the calculated operating resistance and shall size the positive and negative cables and voltage rating of the T/R such that the "calculated" operating output of the T/R complies with all of the following:
- The target commissioning current shall be achieved at a voltage between 30% and 70% of the T/R rated voltage output.
 - The normal operating output current shall be achieved with the voltage adjustment set at more than 10% of the available (rated) T/R voltage output.

Note:

The above section is a mandatory design requirement but is not a mandatory requirement for commissioning acceptance.

- 5.2.5.4** The CP system "operating" circuit resistance measured during commissioning of a new CP system shall not be greater than 90% of the CP power supply "rated" circuit resistance unless all of the following are met:
- Structure is cathodically protected and meeting protection criteria requirements as per this standard.
 - If the CP system power supply is a tap adjustable T/R, it shall have at least two fine tap setting increments remaining.
 - If the CP system power supply is automatic and/or phase controlled type, it shall have at 15% of rated voltage still remaining for further adjustments when required.
 - Each anode current, for the required minimum number of anodes is at or below the maximum commissioning current rating (Table 5 of SAES-X-600).
 - For anode types not mentioned in Table 5 or other types of anode (GRID), the anode operating current shall never exceed the permitted maximum as per design and/or the manufacturer anode data sheet.

5.2.6 Soil Resistivity Surveys

Ensure that the readings are taken for the planned installation depths of the anodes based on the finished grade levels at the plant.

5.2.6.1 Distributed Anode Beds

- Conduct soil resistivity or soil conductivity measurements over a representative sample of the distributed anode installation locations.
- Ensure that the measurements are representative for the planned installation depths of the anodes based on the finished grade levels at the site.
- Show the location that each measurement was taken on a detailed CP layout drawing.
- For plant plots that exceed 50 m by 50 m, soil resistivity survey at multiple locations shall be carried out and the worst case values (lowest resistivity values) used in the anode voltage gradient calculations.

5.2.6.2 Surface Anode Beds

- Conduct soil resistivity or soil conductivity measurements at 10 meter intervals over the full length of the proposed surface anode bed location.
- The four pin Wenner method may be used in areas with nominal soil resistivities below 10,000 ohm-cm. A non-contact electromagnetic soil resistivity/conductivity instrument shall be used where equipment capability favors depending upon the depth and resistivity of the area.
- Measurements shall be taken for the planned installation depth of the anode bed.
- Mark the selected location of the surface anode bed with wooden or metal stakes at each end of the proposed anode bed location. Show the location on a detailed site sketch, and include this information on the CP design drawings.

5.2.6.3 Remote Deep Anode Bed:

In consultation with the Groundwater Protection Unit of the Saudi Aramco Environmental Protection Department, determine the final bore hole depth and anode placement based on the drill stem resistance and test anode resistance measurements collected while drilling the anode hole.

- a. Conduct soil resistivity or soil conductivity measurements for deep anode bed locations.
- b. Anodes to be installed deeper than 15 meters shall be classified as “deep” anode beds, and hence will require the drilling depth to be pre-approved in writing by the Groundwater Protection Unit of the Saudi Aramco Environmental Protection Department.
- c. Record drill stem and test anode resistance measurements and submit for review and analysis to Saudi Aramco (or designate).
- d. The drill stem and test anode measurements shall be taken in accordance with the requirements detailed in Standard Drawing AA-036385. Final acceptable borehole depth anode distribution based on the drill stem and test anode analysis.
- e. Mark the selected location of the deep anode bed with a wooden or metal stake. Show the location on a detailed site sketch, and include this information on the CP design drawings.

5.2.7 DC Power Supply

5.2.7.1 For hazardous areas (maximum Class I, Zone 2), the design agency shall select a cathodic protection DC power supply (and other CP system equipment) that complies with the requirements of NEC Articles 500 to 504 for hazardous (Class I, Zone 2) areas. CP equipment shall not be placed in Class 1 Zone 1 areas.

5.2.7.2 Install the rectifiers in non-hazardous areas where possible and use oil-immersed rectifier units inside hydrocarbon plant areas, in all locations that are within 30 meters of hydrocarbon plant perimeter fencing (outside), and within 1 km of a coastline.

5.2.7.3 Designs shall be based on T/Rs manufactured in compliance with 17-SAMSS-004 or 17-SAMSS-005.

Exception:

The DC power supply for a grid or continuous impressed current anode CP system can be air cooled (NEMA 4X) and need not comply with the requirements of 17-SAMSS-004 and 17-SAMSS-005 but shall be supplied by a Saudi Aramco approved manufacturer.

5.2.7.4 Install the CP power supplies in accordance with Saudi Aramco Standard Drawing AB-036378 sheets 1 and 2.

5.2.7.5 Use air cooled T/Rs (type AA) for indoor or non-classified, non-marine environment applications. T/Rs with NEMA Class 3R enclosures shall NOT be used inside hydrocarbon plant areas or in marine environments unless installed indoors.

5.2.7.6 DC power supplies shall have a maximum rated output voltage of no greater than 100 volts. The sizing of the T/R shall be optimized based on the circuit resistance in accordance with Section 5.2.5 of this standard.

5.2.7.7 Transformer/Rectifiers and junction boxes (as per 17-SAMSS-008) shall be permanently and systematically labeled in line with the plant electrical devices and equipment labeling system. Use T/R for transformer/rectifier and J/B for junction box. Transformer/rectifier example is ANDR-GOSP-1-T/R-07. Junction box example is ANDR-GOSP-1-T/R-07-J/B-12.

5.2.7.8 CP Power Supply Rating

CP systems shall be designed with CP power supplies sized with a minimum rated output current equaling the sum of the current requirement plus a design surplus of 30%.

Note:

When selecting the current rating for power supply units, if the use of 30% surplus results in a rating of say 104 or 105 amps whereas the next standard power supply rating available is 150

amps. It is economical to use 100 amps which will still give surplus of more than 20%. Therefore, use of surplus capacity must be carefully selected ensuring a minimum surplus of 20%.

5.2.7.9 Standard/Recommended CP Rectifier Rating

For the purpose of cathodic protection, rectifier ratings must be selected from the standard list in Table 6.

5.2.8 Anodes

5.2.8.1 Unless specified otherwise in this document, impressed current anode materials for soil applications shall be high silicon cast iron or mixed metal oxide complying with 17-SAMSS-007 and shall use design parameters as detailed in Table 4.

5.2.8.2 Galvanic anode materials for soil applications shall be manufactured in accordance with 17-SAMSS-006. Designs using galvanic anodes shall comply with the design parameters detailed in Table 3.

5.2.8.3 In asphalt, paved areas, or tightly compacted marl areas, conventional impressed current anodes shall be vented according to Standard Drawing AA-036346. Vent openings shall be above-grade if product spill could contaminate the anode backfill.

5.2.8.4 Use zinc anodes in lieu of magnesium anodes where the soil resistivity is below 500 ohm-cm.

5.2.9 Cables

5.2.9.1 Cathodic protection cables shall be manufactured in accordance with 17-SAMSS-017.

5.2.9.2 The minimum size for the primary positive and negative DC cables from the rectifier shall be 25 mm² (#4 AWG).

5.2.9.3 All cathodic protection cables including the primary positive and negative cables shall be sized to comply with the most recent edition of the National Fire Protection Association NFPA 70, National Electric Code (NEC).

5.2.9.4 Unless otherwise specified by the cable manufacturer, the allowable ampacity of High Molecular Weight Polyethylene (HMWPE) cables manufactured in accordance with 17-SAMSS-017 shall be rated for ampacity under the column for insulation rated for 90°C in the NFPA 70, NEC Handbook. Correction shall be made to an operating conductor temperature of 40°C.

5.2.10 Structure Electrical Continuity

5.2.10.1 Install dedicated bonds to all structures influenced by the impressed current anode systems to ensure electrical continuity back to the cathodic protection rectifier including metallic valve flange connections with or without rubber gaskets (mechanical joints are not reliable joints for electrical continuity). Make all cable connections to structures according to Standard Drawing AB-036381.

5.2.10.2 Cables installed for, or associated with electrical grounding or electrical ground continuity purposes shall not be used to achieve the required electrical continuity for cathodic protection.

5.2.10.3 Identify and locate all bonding connections on the cathodic protection drawings.

5.2.11 Resistors for Current Distribution Control

Electrical resistors shall not be used for current distribution control.

5.2.12 Bonding

5.2.12.1 All buried metal structures influenced by cathodic protection systems shall be electrically continuous. Install continuity bonds between all buried structures requiring cathodic protection that are not welded together including metallic valve flange connections with or without rubber gaskets (mechanical joints are not reliable joints for electrical continuity).

A separate reliable connection of the anode to the valve either by a bond cable across the flange or by a separate cable from each metallic part of the valve to the test point shall be

provided to ensure proper electrical continuity. Make all bond cable terminations between structures in an above-grade enclosure constructed according to one of the following Standard Drawings AA-036276, AA-036277, AA-036349, AA-036350, and AA-036637. Make bond connections for manholes according to Standard Drawing AA-036637.

5.2.12.2 The bond conductor shall be sized to achieve a voltage drop across the conductor of less than 0.05 volt and shall be minimum 16 mm² (#6 AWG).

5.2.12.3 Make all cable connections to structures according to Standard Drawing AB-036381.

5.2.12.4 Install a test lead on all buried bonded structures in addition to the bond cable. Terminate the test lead in the bond box on its own individual terminal.

5.2.12.5 If an above ground bond box will interfere with the operation of the plant, a flush mounted bond box installed according to Standard Drawing AA-036637 may be used provided the cathodic protection proponent organization concurs in writing.

5.2.13 Electrical Isolation

5.2.13.1 Isolating flanges, isolating joints or spools, or other electrical isolating devices are not allowed except to limit protective current flow between plant facilities and off-plot pipelines, and for short pipeline sections less than 500 meters long protected by a galvanic anode system.

Exceptions:

Fire water systems at office buildings may be electrically isolated.

Isolation flanges may also be used to electrically isolate protected and non-protected piping sections to prevent loss of CP current.

Note:

Purpose of section 5.2.13.1 is to ensure that all CP protected structures within the plant are treated as a common cathode to prevent stray current interference.

5.2.13.2 For hazardous locations where if use of isolation devices are prevented for safety reasons, then supplemental cathodic protection shall be provided at these locations to ensure proper protection.

5.2.13.3 Where the installation of electrical isolation is required for the purposes of CP, then all possible piping pathways including the main pipe, branches, instrumentation lines shall all be considered for the purpose of electrical isolation.

5.2.13.4 For details of pipeline isolation, isolation devices and tests see NACE SP0286 & ISO-15589-1.

5.2.13.5 Type of isolation joint is not dictated by this standard and is based on pipeline size, service requirements such as temperature, pressure, chemical resistance, mechanical properties. Use of insulating unions, flange type isolation devices, monolithic isolation devices as well isolation spools are all considered appropriate. However, for wet services where the internal fluid may be electrically conductive, use of isolating spool pieces are considered more appropriate unless it can be evidently demonstrated with the help of internal lining and/or coating that flange type connection can be effective.

5.2.13.6 Electrical isolation devices if installed, shall be protected from lightening and surges by surge diverters / spark gaps/ solid state polarization cells. Such devices must be fully encapsulated and suitable for installation in hazardous areas. Their voltage and current rating shall be based on expected surges and capable of safely discharging the expected surge/lightning currents without any damages and/or breakdown. Such devices must be regularly monitored and maintained for their integrity. Any failed devices must be replaced immediately. Cathodic Protection design documents shall provide a detailed section for selection of devices as stated above along with a detailed monitoring and maintenance procedure.

5.2.13.7 Do not install isolating devices in any buried or submerged portions of a pipeline.

5.2.13.8 Provide an above-grade bond box for all isolating devices. Construct the box according to Standard Drawings AA-036277, AA-036349 or AA-036350.

Exception:

An above grade bond box is not required for isolation devices mentioned in Section 5.2.

5.2.14 Hazardous Areas

5.2.14.1 For CP equipment and materials such as T/R unit and Junction boxes, as much as possible these must be located in non-hazardous areas. However, if these are to be located in hazardous areas (maximum Class I, Zone 2), the design agency shall select a cathodic protection DC power supply (and other CP system equipment) that complies with the requirements of NEC Articles 500 to 504 for hazardous (Class I, Zone 2) areas.

5.2.14.2 Plot plan with hazardous area overlay drawing showing clearly all structures that are CP protected and all CP equipment and their locations shall be provided.

5.3 Buried Metal Pipelines

5.3.1 Design Philosophy

The cathodic protection system for a buried metal pipeline shall be designed using the "earth potential rise" method with distributed impressed current anodes or continuous linear anodes and the relative equations detailed in Appendix I.

5.3.2 Protection Criteria

The negative pipe-to-soil potential shall be:

- A minimum of -1.0 volt and a maximum of -2.5 volts, with reference to a copper/copper sulfate electrode and the current on, or,
- If all of the protective current can be interrupted, or a corrosion coupon or cathodic protection assessment probe is used, a minimum potential of either -0.850 volt off with reference to a copper/copper sulfate electrode, or
- a 100 mV polarization shift between the instant off potential and either the native state potential (for initial commissioning only) or the depolarized state potential.

Notes:

- Criteria a. (ON Potential) Maximum ON potential limit stated here refers to potentials with all influencing CP power supplies turned ON and if with the help of a coupon or other reliable method if it can be demonstrated that the instant OFF values are within the acceptable limits and in such specific cases, higher ON potentials than the limit stated will be allowed.*
- Criteria c. (Polarization Potential) The "potential decay" is the decrease in potential between the "instant off" and the steady state off potential. A probe or coupon typically reaches a steady state off potential within 60 seconds to 60 minutes. Refer to ISO15589-1 and NACE TM0497 for depolarization measurement procedures. This criteria shall not be used when mixed-metal components or dissimilar metals are in contact for example steel pipeline in contact with copper (pipe or earthing). And in soils containing Sulfate Reducing Bacteria (SRBs), or when interference or stray current activity is present.*
- Protection criteria stated in here applicable to buried and submerged carbon steel, , cast iron, ductile iron materials only.*
 - When specials stainless steel and other corrosion-resistant alloys are used which are prone to SCC, more positive potentials than what is stated here may be applicable.*
 - Similarly for other materials like copper, aluminum, stainless steel and other alloys different criteria will be applicable (Refer to ISO 15589-1 & NACE standards).*

4. *All pipe-to-soil potentials measured with reference to a Cu/CuSO₄ reference electrode. Dissimilar metal piping requires that the polarized potential be maintained on all surfaces of the structure based on the protection requirements for the most anodic metal. For pipelines in anaerobic soils in which there are significant quantities of SRBs, the potential values indicated shall be increased by further 100 mV. (For example when using minimum off potential values, it shall more negative than -950 mV CSE).*

5.3.3 DC Power Supply

The DC power supply shall comply with Section 5.2.7 of this standard.

5.3.4 Anode Beds

- 5.3.4.1 The number and size requirements for the anodes shall be determined based on providing a sufficient potential gradient over the entire length of the respective pipeline to achieve the protection criteria.
- 5.3.4.2 The resistance to ground of a distributed anode bed does not need to compliment the rated resistance of the rectifier, however, the combined current capacity of the anodes shall be equal to or greater than the rectifier rated current.
- 5.3.4.3 Anode spacing shall be considered based on the voltage gradient calculations. It is required to ensure that the recommended anode quantity and spacing covers the entire piping network that is part of the CP system. One good approach to satisfy this requirement is to verify on the plot plan with known area, anodes are spaced enough based on voltage gradient and then draw a circle of gradient for each anode to show that these gradient circles overlap each and other and covers the entire piping under consideration. 'A0' sized plot plan drawing showing anode layout, anode voltage gradient circles (anode voltage cones) and UG piping shall be provided with the design for review.
- For plant plot that exceed 50 m by 50 m, soil resistivity survey at multiple locations shall be carried out and the worst case values (lowest resistivity values) used in the anode voltage gradient calculations.
- 5.3.4.4 If impressed current anodes are placed within 50 meters of a thrust anchor, then two additional impressed current anodes shall be installed within 15 meters of the anchor, and placed on opposite sides of the anchor. If impressed current anodes are installed, galvanic anodes are not required for the thrust anchor.
- Note:*
- Cathodic protection requirements for thrust anchors on pipelines exiting or entering the plant are covered in SAES-X-400.*
- 5.3.4.5 Galvanic anodes may be used to protect short (less than 500 meters) buried sections of coated piping; provided the following conditions are met:
- All parts of the piping are greater than 100 meters from the closest impressed current rectifier.
 - The piping is not shielded by other structures and no other structures (concrete, plastic, wood, and metal) are within 6 meters of the piping.
 - The piping is not electrically continuous with any other buried uncoated structure.
 - The cathodic protection design shall be based on calculation of the current requirement for the short section of buried pipeline, and determination of the required number and size of galvanic anodes to achieve that current for the minimum life requirement.
 - For multiple short sections of piping less than 500m each, the combined length shall be less than 500m. Otherwise, the preferred method in such cases shall be ICCP system considering common cathode to avoid multiple isolations in a plant facility.

5.3.5 Monitoring

- 5.3.5.1** Provide a test station according to Standard Drawing AB-036907 for measuring pipe-to-soil potentials at:
- Insulated cased crossings
 - Paved road crossings
 - Thrust-bored road crossings
 - Negative connections to underground piping
 - Buried bond connections
 - Other locations as required by operational needs.
- 5.3.5.2** The maximum spacing between test stations or risers used for test stations is 100 meters.
- 5.3.5.3** Remote Monitoring Test Stations as per section 5.1.2.4 shall be provided.
- 5.3.5.4** 25% of test stations shall include CP coupons and these be strategically deployed at drain points (DPs) and most distant pipe locations from DPs.
- 5.3.5.5** Piping risers may be used for test stations provided they are labeled and identified as test points on the Cathodic Protection Layout Drawings and on the riser.
- 5.3.5.6** Install soil access test holes, according to Standard Drawing AB-036907, in concrete and asphalt (paved) areas directly over the top of the pipe or structure to be monitored. In these concrete or asphalt areas, install one soil access hole at each of the following locations:
- Midway between anodes.
 - One over the structure closest to each anode.
 - One at the beginning and one at the end of each pipeline segment.
 - One at each metallic hydrant, metallic riser or metallic valve that is part of a non-metallic piping system. However, if there is sufficient soil access at the base of the hydrant, riser or valve, a soil access test hole is not required.
 - One at each test station or riser used as a test station.
- 5.3.6** **Bonding, Electrical Continuity, and Electrical Isolation**
Bonding, electrical continuity, and electrical isolation shall comply with Section 5.2.12 of this standard.
- 5.3.7** **Buried Metal Pipelines - Temporary Cathodic Protection**
- 5.3.7.1** Temporary CP of plant piping sometimes impractical due to arrangement and construction; therefore, requirement for temporary CP shall be carefully evaluated at the project design stage. Various factors that needs to be considered for such evaluations include but not limited to following:
- Soil corrosivity
 - Presence of copper earthing network
 - Other corrosion protection measures such as coating, corrosion allowance considerations
 - construction period
- 5.3.7.2** If the permanent ICCP will not be energized within 6 months of burial of any portion of the pipeline, then temporary CP shall be provided.
- 5.3.7.3** The current density requirements for temporary CP shall be 10% of current density stated in Table 2 of this standard.
- 5.3.7.4** The design life of temporary cathodic protection shall be 2 years minimum.
- 5.3.7.5** Instead of separate temporary CP system, permanent CP installations can be temporarily energized to achieve temporary protection.
- 5.3.7.6** Where supplemental galvanic anodes are determined to be necessary for temporary cathodic protection, wherever practical, they shall be installed at permanent galvanic anode locations.

The permanent junction box or test station facilities shall be used to terminate the supplemental galvanic anodes.

5.3.7.7 For temporary protection whilst there is no need to apply specific protection criteria, pipe-to-soil potential survey data as well as current from the anode to connected piping will be required to verifying that the temporary system is providing current to the piping.

5.3.7.8 At locations where the permanent galvanic anode equipment will not work for effective temporary cathodic protection:

- Select low resistivity locations to maximize the current output and minimize the number of anodes required.
- Install galvanic anodes on an "as required" basis, i.e., use the calculated numbers of anodes as a guide, and install anodes as required basis to ensure that the connected piping are receiving current.

5.3.7.9 Terminate galvanic anodes for temporary cathodic protection in a 3-pin test station using the installation details given in Standard Drawing AA-036352. The number of anodes required per location shall be based on the actual requirement and not as shown on the standard drawing.

5.3.8 Cathodic Protection of existing unprotected Pipelines

When it is desired to cathodically protect existing unprotected structures, considering that these are existing pipelines whose condition is unknown, it is recommended to carry out a current injection test. Current injection test in addition to providing current requirement for CP, also verifies the capability of cathodic protection system to protect the structure of interest. Such current injection tests can be carried out utilizing temporary DC power sources (including batteries with variable resistors), temporary anode beds (usually made of scrap steel) to determine current required for cathodic protection and probable locations for anode beds.

For existing structures prior to design of CP system, other asset integrity tests such as physical condition assessment (sample dig survey), fitness for purpose, leak and/or failure history, etc..., shall be assessed. When cathodically protecting existing unprotected structures, it shall be brought to the notice of all concerned including the proponent that effective CP system whilst providing protection from further corrosion does not correct any pre-existing corrosion which may still result in accelerated corrosion and/or failure and hence over reliance on CP under such conditions shall be avoided unless all such pre-existing corrosion defects have been identified and repaired.

5.4 Isolated Buried Pipeline Casings

5.4.1 Design Philosophy

The cathodic protection system for a buried metal casing installed for a buried section of a metal pipeline shall be:

- a. Designed using the "earth potential rise" method for casings of normally buried metal pipelines protected by impressed current anodes, or,
- b. designed using galvanic anodes for casings of normally above grade pipelines that are cased for crossings such as at roads or dikes.

5.4.2 Protection Criteria

Refer to section 5.3.2

5.4.3 Anodes for Pipeline Casings

5.4.3.1 Galvanic Anodes

5.4.3.1.1 Install the number of anodes specified in Table 7. Use 27.2 kgs (60 lbs) magnesium anodes for soil resistivities greater than 500 ohm-cm. Use 22 kgs (48.5 lbs) or 50 kgs (110.23 lbs) zinc anodes for soil resistivities equal to or less than 500 ohm-cm.

5.4.3.1.2 Galvanic anodes shall be connected to the casing through galvanic anode (3-pin) junction boxes (17-SAMSS-008). See Table 7.

5.4.3.1.3 For casings not addressed by this table, such as bare casings or casings longer than 30 meters, a site-specific design must be completed that considers soil resistivity, total current requirement, and anode life. The supervisor of the Cathodic Protection and Coatings Unit must concur with the proposed design.

5.4.4 Monitoring

Monitoring shall comply with Section 5.3.5 of this standard.

5.4.5 Bonding and Electrical Continuity

5.4.5.1 Bonding and electrical continuity, shall comply with sections 5.2.12, and 5.2.10 of this standard.

5.4.5.2 At crossing locations involving more than one casing, the casings shall be bonded together.

5.5 On Grade Tank Bottoms - Soil Side

5.5.1 New Tanks - CP Design Philosophy

5.5.1.1 The cathodic protection system shall be designed in accordance with the requirements for new tanks detailed in Standard Drawing AA-036355.

5.5.1.2 New tanks shall be designed with an impressed current grid system, loop system, or continuous distributed anode system placed directly beneath the tank (see 5.5.1.4) and designed to ensure a relatively uniform current distribution that maintains all points on the tank bottom within the required protection criteria limits.

Note:

In case of a tank(s) containing high temperature product(s) typically in the region of 90°C and above that could make the soil environment under the tank bottom to be very dry making the CP system ineffective, then there is no need for a CP system to protect the tank(s) bottom. However, this commentary note is only applicable when the said tanks are operating at high temperatures. If the said tank is used for some other purpose (lower operating temperatures) or operated at temperatures much lower than 90°C then cathodic protection will be required. Therefore, decision to include or exclude CP shall be case-to-case evaluation for new tanks as applicable.

Before any decision can be made to exclude CP, CONTRACTOR are to provide the temperature profile through the appropriate heat transfer calculations that include coating skin temperature on the underside of the tank base, soil temperature directly on the underside of the tank base, the soil temperature a vertical distance 500 mm below the tank base where anodes and cables would potentially be installed.

The heat transfer calculations will involve specifying tank design and operating parameters including (but not limited to) the following:

- Tank operating temperature envelope
- Coating types (internal and external)
- Thicknesses and coating heat transfer values
- Tank base thicknesses and steel heat transfer values
- Ambient soil temperatures and soil heat transfer values

When it is deemed necessary to install cathodic protection for such high temperature application, then all the CP materials such as anodes, reference electrodes, coupons and corrosion probes, cables, connections, etc..., must be suitably rated and/or corrected for the temperature of installation.

5.5.1.3 The backfill sand for the tank external bottoms shall be clean washed chloride free sweet sand (Bitumen Sand / Oily Sand NOT permitted). (Refer to 32-SAMSS-005 Appendix B for details)

- 5.5.1.4** Tank bottoms are usually provided with secondary containment liners as dictated by SAES-D-100, primarily for protection of the soil from contamination by the stored product during leak. CP considerations differ for the tanks with and without secondary containment liners.
- For the tanks with liners, anodes and monitoring facilities such as reference electrodes, coupons and ER probes etc. are installed within the liner directly under the tank bottom. Current drain to other nearby structures such as grounding are negligible and hence ignored.
 - For the tanks without any secondary containment liners, anodes can be directly installed under the tank bottom or distributed type or any other acceptable configuration. Monitoring facilities such as reference electrodes, coupons and ER probes etc. are installed directly under the tank bottom. In such cases current drain to nearby structures such as earthing, and rebars, etc., must be factored into the design.
- 5.5.1.5** For new tank external bottoms, a soil resistivity value of 50,000 ohm-cm can be used for calculations unless the actual field verified values indicate differently.
- 5.5.1.6** **Monitoring**
- 5.5.1.6.1** Monitoring shall comply with Section 5.3.5 of this standard.
- 5.5.1.6.2** Perforated/slotted monitoring tube installed across the center of the tank bottom will be required to monitor tank bottom potentials. Appropriate purpose made reference electrodes with moist backing (e.g., wet foam) shall be used to pull through the perforated monitoring tube for the purpose of tank to soil potential measurements. Flooding of tank bottoms is not permitted.
- 5.5.1.6.3** Stationary reference electrodes & CP Coupons capable of working in tank bottom installations shall be provided for monitoring at selected locations.
- 5.5.1.6.4** ER probes capable of working in tank bottom installations shall be provided for monitoring at selected locations.

Note:

ER probes are provided for regular ongoing monitoring and not for the purpose of commissioning. ER probe data shall be collected and recorded at the time of commissioning for review later (see also Section 5.5.8).

- 5.5.1.7** For the tank bottoms with GRID type systems, number of power feed connection to the anode GRID shall be based on Table 8.

5.5.2 **Existing Tanks - CP Design Philosophy**

It is up to the proponent organization to evaluate the corrosion history of the tank(s) and to conduct the required field measurements to facilitate a justifiable engineering decision on the need to replace or upgrade a cathodic protection system. Appendix II contains a description of field measurement procedures that can be used to determine if CP will be effective for tank bottoms on oily sand.

If the decision is to install a cathodic protection system, then the following procedure shall be implemented:

- a. Conduct soil resistivity measurements approximately 10 meters from the tank to determine uniformity of the soil surrounding the tank and to determine the optimum depth and orientation to place the anodes.
- b. Install one test anode remote from the tank (25 to 50 meters) at the depth determined by step a) above and connect a DC power supply between the anode and the tank set to discharge the rated current from this anode.
- c. Measure the resulting gradient to determine the optimum distance the anodes shall be place from the tank.
- d. Design and install the anodes circumferentially distributed around the tank based on the information gathered from the above testing. Conduct this field testing for each group of tanks within the same plant sharing similar soil characteristics.

5.5.2.1 The cathodic protection system shall be designed in accordance with the requirements for existing tanks detailed in Standard Drawing AA-036355.

5.5.2.2 The anode configuration for existing tanks requiring new impressed current cathodic protection systems, or upgrades to existing impressed current cathodic protection systems shall be designed using impressed current distributed anodes placed around the perimeter of a tank.

Note:

Existing tanks that require anode upgrades and have an oily sand foundation mixture or asphalt or concrete foundation pad will not receive complete cathodic protection in areas where the oily sand mixture collects water at the interface between the tank bottom and the oily sand. Consideration shall be given to removing the oily sand wherever economically viable.

5.5.2.3 Existing tanks requiring new tank bottoms that will be installed above the existing bottom without removal of the existing bottom shall be cathodically protected with galvanic anodes in accordance with Standard Drawing AA-036905.

5.5.3 Protection Criteria - Tank Bottoms and Buried Tanks

For new tank bottom CP system installation, the negative structure-to-soil potential shall be measured at 1 meter intervals through a slotted monitoring tube installed beneath the tank, and through access holes in the ring wall. The protection criteria for tank bottoms and buried tanks shall be:

- a. A minimum of -1.0 volt and a maximum of -2.5 volts, with reference to a copper/copper sulfate electrode and the current on, or
- b. if all of the protective current can be interrupted, a minimum potential of either -0.850 volt off with reference to a copper/copper sulfate electrode, or
- c. a 100 mV polarization shift between the instant off potential and either the native state (for initial commissioning only), or the depolarized state potential.

Notes:

1. *Criteria a. (ON Potential) Maximum ON potential limit stated here refers to potentials with all influencing CP power supplies turned ON and if with the help of a coupon or other reliable method if it can be demonstrated that the instant OFF values are within the acceptable limits and in such specific cases, higher ON potentials than the limit stated will be allowed.*
2. *Criteria c. (Polarization Potential) The "potential decay" is the decrease in potential between the "instant off" and the steady state off potential. A probe or coupon typically reaches a steady state off potential within 60 seconds to 60 minutes. Refer to ISO15589-1 and NACE TM0497 for depolarization measurement procedures. This criteria shall not be used when mixed-metal components or dissimilar metals are in contact for example steel pipeline in contact with copper (pipe or earthing). And in soils containing Sulfate Reducing Bacteria (SRBs), or when interference or stray current activity is present.*

5.5.4 DC Power Supply - Tank Bottoms

The DC power supply shall comply with Section 5.2.7 of this standard.

5.5.5 Anodes - Tank Bottoms

5.5.5.1 For an existing tank, HSCI, mixed metal oxide or polymeric anodes shall be used for existing tank designs.

5.5.5.2 Mixed metal oxide, or polymeric anodes shall be used for new tank designs. The design concept and the anode specifications and manufacturer must also be pre-approved in writing by Saudi Aramco.

5.5.5.3 Impressed current anodes for existing tanks, and galvanic anodes shall comply with Section 5.2.8.

5.5.5.4 The resistance to ground of the anodes can be greater than the rated resistance of the rectifier, however, the combined current capacity of the anodes shall be equal to or greater than the rectifier rated current.

5.5.6 Monitoring

5.5.6.1 For a new tank bottom, monitoring shall be through access tubes through tank ring walls, stationery reference electrodes, CP coupons, corrosion probes and slotted monitoring tubes under the tank bottom according to Standard Drawing AA-036355.

5.5.6.2 For an existing tank with distributed anodes outside the perimeter of the tank, monitoring shall be through access tubes through the ring wall, as per Standard Drawing AA-036355, Detail 6.

5.5.7 Bonding and Electrical Continuity

5.5.7.1 A dedicated negative cable shall be connected directly between each tank protected and the rectifier providing current to that tank. Connect multiple tanks to the rectifier through a negative junction box.

5.5.7.2 For large diameter tanks with transformer rectifier rating of 75 Amps and more, at least two dedicated negative connections to each tank shall be provided.

5.5.8 Commissioning

5.5.8.1 Tank external bottom commissioning shall be carried out with the tanks filled to at least half of its capacity. This is required to achieve reasonable tank to soil contact. If necessary for the purpose of commissioning water shall be filled or the commissioning carried out at the time of hydro test / water fill up test.

5.5.8.2 Each tank must be provided with at least two ER probes in accordance with AA-036355. Provision shall be made at the ER probe test station to connect and/or disconnect all of the ER probes to the CP system through a switch. For the first year after commissioning the center ER probe shall be connected to the CP system to evaluate status of protected probe.

5.5.8.3 Initial set of ER probe data shall be collected and recorded within 48 hrs of installation of the probes. Second set of ER probe data shall be collected just before commissioning of the CP system and at least once a month for the first 12 months. Thereafter, the monitoring frequency can be reduced to once every 3 months to evaluate the corrosion status. ER probes data is not used to decide on the CP system status at the time of commissioning.

5.6 Buried Valves and Fittings

5.6.1 Design Philosophy

Cathodic protection for small buried isolated metallic structures that are part of a non-metallic piping system, such as valves and risers shall be protected using galvanic anodes, in accordance with Standard Drawing AA-036629.

Exception:

For buried metallic fittings used in non-metallic firewater service or any other water service (other than hydrocarbon), a supplemental (additional) coating system using APCS-113c over top of the original shop or factory applied coating system can be applied as an alternative to galvanic anodes. Metallic valves and risers are not covered under this exception.

5.6.2 Protection Criteria

The design of the cathodic protection system shall achieve a minimum structure-to-soil potential for isolated buried metal structures of -1.0 volt with reference to a copper/copper sulfate electrode and the current on, except if double coatings are used in lieu of CP as detailed in 5.6.1 Exception above.

5.7 Metal Piles, Sheet Piling and Associated Anchors

5.7.1 Design Philosophy

Design cathodic protection for metal piles using impressed current (remote or distributed anode system) or galvanic anode systems. The current requirements for all metallic buried structures that are electrically continuous with the protected structure must be considered.

5.7.2 Protection Criteria

The minimum structure-to-soil potential for design purposes shall be:

- a. Water: The structure-to-water potential for sheet piling shall be a minimum of -0.90 volt, with reference to a silver chloride electrode and the current on.
- b. Soil: The structure-to-soil potential for piling shall be a minimum of -0.85 volt, with reference to a copper/copper sulfate electrode and the current on.

5.7.3 DC Power Supply

If an impressed current cathodic protection system is used, the DC power supply shall comply with Section 5.2.7 of this standard.

5.7.4 Bonding and Electrical Continuity

Impressed current designs for pile systems require electrical continuity between all piles within a common structure, electrical continuity by welded rods or brazing of cables (mechanical joints are not reliable joints for electrical continuity).

6 Installation, Records, Commissioning, and Inspection

Refer to Saudi Aramco Best Practice SABP-X-003. SABP-X-003 shall be deemed a mandatory document for this Standard.

Table 1: Potential Criteria

Equipment Surface Description	Section
Buried Metal Pipelines	5.3.2
Isolated Buried Pipeline Casings	5.4.2
On Grade Tank Bottoms and Buried Tanks	5.5.3
Buried Valves, Hydrants, Monitors and Fittings	5.6.2
Metal Piles, Sheet Piling & Associated Anchors	5.7.2

Table 2: Minimum Surface Current Densities

Equipment Surface Description	Current Density (mA/m ²)
Bare Copper Ground Rods ⁽³⁾	50
Stainless Steel Ground Rods (driven)	20
Bare Tinned Copper Cable ⁽³⁾	70
Bare Copper Cable ⁽³⁾	100
Bare Steel (buried) ⁽¹⁾	20
Polyethylene Tape coated steel ⁽²⁾	1.25
Liquid Applied Epoxy coated steel ⁽²⁾	0.75
Fusion Bond Epoxy coated steel ⁽²⁾	0.10

Note (1): For tank bottom external CP calculation, consider the tank bottom bare even if coated. This is to account for the expected coating damages during tank bottom construction. No further safety factor or further current drainage allowances required for Surface Area, Temperature, Micro-Biological corrosion, etc..

Note (2): Current density specified above is applicable for ambient temperatures up to 30°C. For structures operating at higher temperatures, suitable temperature correction for current density will be required.

Note (3): For copper materials used in earthing and other applications, the current density values to polarize significantly increases in aerated soils. Therefore, when soil conditions indicate significantly high resistivities i.e. higher than 50000 ohm-cm, the current density values indicated here shall be doubled to achieve CP polarization.

Table 3: Galvanic Anode Material Properties

Anode Material	Consumption (kg/A-Y)	Potential (mV) (Cu/CuSO ₄)
Aluminum	3.8	-1,100
Magnesium	10.95	-1,700
Zinc	11.8	-1,100

Note: The consumption rates detailed in Table 3 include efficiency; however, a utilization factor of 0.85 may be included for design purposes at the designer's discretion

Table 4: Impressed Current Anode Design Parameters

Anode Material	Consumption (kg/A-Y)	Current Density Maximum (mA/cm ²) ²
HSCI	0.45	0.7
Mixed Metal Oxide (MMO)	MMO Anodes shall comply with requirements mentioned in Section 7.1.3 of 17-SAMSS-007	

Note: The consumption rates detailed in Table 4 have been corrected for efficiency and utilization factor. Shortening of the effective anode length for a HSCI anode due to an end cap at the cable end of the anode shall be neglected in the theoretical calculation of anode current capacity.

Table 5: Impressed Current Anode Data

Type	Dimensions	Weight (kg)	Current Capacity per Anode (Amps)	Maximum Commissioning Current (Amps)
TA-2	56 mm x 2133 mm	20.5	2.63	4.0
TA-4	95 mm x 2133 mm	38.6	4.45	7.0
TA-5A	121 mm x 2133 mm	79.4	5.67	10.0
Tubular MMO	16 mm x 1,200mm	n/a	4.2	6.0
Tubular MMO	19 mm x 1,200mm	n/a	5.0	7.0
Tubular MMO	25 mm x 1,200mm	n/a	6.5	9.0

Notes:

- 1. The maximum acceptable current during commissioning is based on manufacturer's ratings and shall not be used for design. This number shall only be used when determining commissioning acceptance.*
- 2. The consumption rates detailed in Table 5 include utilization factor and effective anode shortening due to end caps.*
- 3. The Tubular MMO anodes stated above are typical and other anode dimensions may be selected as required based on the sizes available in the market.*

Table 6: Standard CP Rectifier Rating

Voltage rating (volts)	10 / 20 / 30 / 40 / 50 / 60 / 100
Current rating (Amps)	5 / 10 / 15 / 25 / 30 / 50 / 60 / 75 / 100 / 150

Table 7: Galvanic Anodes for Coated Casings that are Electrically Isolated

Casing Diameter (inches)	Number of Casings	Casing Length (meters)	Number of Anodes
Any casing diameter	1 – 3	Less than 15	2 on one end
Any casing diameter	1 – 3	15 to 30	2 on each end

Notes:

- 1. The above table is for groups of casings that are spaced no more than 6 meters from each other.*
- 2. For groups of casing greater than 3, divide the number of casings by 3 and multiply the number of anodes in the table by the result of this division. Round up to the next whole number for fractional results.*

Table 8: Number of Power Feed Connections

No.	Current Rating (Amps)	Minimum Number of Power Feeds
1	5 / 10	3
2	15	4
3	25 / 30	5
4	50	7
5	60 / 75	10
6	100	15
7	150	20

Notes:

- 1. Minimum of three power feed connections.*
- 2. At least one power feed connection must be provided on the perpendicular conductor bar.*

Appendices

Appendix A – Earth Potential Rise Formulas

For a Single Vertical Anode

$$\Delta V_x = \frac{0.5 \times I \times \rho}{\pi \times L} \left[\ln \frac{\sqrt{L^2 + x^2} + L}{x} \right]$$

For a Single Horizontal Anode

$$\Delta V_x = \frac{I \times \rho}{\pi \times L} \left[\ln \frac{\sqrt{(0.5L)^2 + x^2 + h^2} + 0.5L}{\sqrt{x^2 + h^2}} \right]$$

Where:

- ΔV_x = Earth potential change (volts) at distance “x” from the anode
- I = Current flow (amperes) from the anode
- ρ = Soil resistivity (ohm-cm)
- L = Anode length (cm)
- x = Horizontal distance (cm) from the anode to a given point on the structure
- h = Depth of burial (cm) to the centerline of the anode

Appendix B – Tank Bottom CP Testing Procedure for Tanks Sitting on Oily-Sand Pads

Trip #1:

Objective:

Conduct soil resistivity measurement to specify optimum remote anode depth that will generate the highest voltage gradient.

CONTRACTOR will:

1. Pick a (remote) location around 10-40 meters away (10 meters for low resistivity areas and 40 meters for high resistivity areas) from existing tank under T&I specified by Proponent, free from any buried metallic structures.
2. Measure remote 4-pin soil resistivity at the proposed remote anode location at depths from 1 to 6 m with 0.5 m pin-spacing intervals, in low resistivity areas conducive to four pin measurements. In high resistivity areas, four pin measurements and Geonics readings will also be taken at 1.5 meters, 3 meters and 6 meter depths.
3. Calculate layer resistivity using the Barnes Two-Layer Method.
4. Determine the best anode installation arrangement (horizontal or vertical and at what depth) as dictated by the highest resistivity layer measured that will accommodate anode length (either horizontally or vertically). Ensure that the entire anode length shall be either all above or all below the water table, where water tables are present.
5. Recommend the optimum orientation (horizontal or vertical), depth, and location for the Proponent to install one remote anode.

Proponent will:

6. Install one TA-4 remote anode as described in point 5 above. Use at least 200 pounds of coke breeze backfill to pour all around the anode.
7. Connect anode to existing or temporary T/R.

Trip # 2:

Objective:

Perform voltage gradient mapping using the remote anode to specify the optimum distances at which the voltage gradient is 1 V and 0.5 V, so as to use this data to specify where to place three close anodes (anode-to-tank and anode-to-anode).

CONTRACTOR will:

8. Energize the anode so that current output is approx. 4 Amps.
9. Map voltage gradient around the anode with 2 RE: (1) one fixed (remote) and (2) one approaching anode. Record the gradient measured at approximately 1 meter intervals beginning 20 meters away from the anode on the side opposite of the anode from the tank and working towards the tank. Continue taking the measurements until you reach the edge of the tank. Take special note of the distances at which the voltage shift measured are 0.5 volt and 1 V on each side of the anode. This will enable CONTRACTOR to determine the optimum anode-to-tank (where voltage is 1 V) and anode-to-anode (where voltage is 0.5 V) spacing that will result in a consistent 1-V shift on the tank bottom as measured from outside the tank.
10. Recommend Proponent to install 3 additional TA-4 anodes around the tank at the depth, distance, and spacing determined by the foregoing tests, as dictated by the result of the voltage gradient mapping conduct on one remote anodes.
11. Specify location of three 1 meter by 1 meter coupons to be cut from the tank bottom plate to facilitate potential measurements from inside the tank. Location of the coupons is determined so as to facilitate obtaining optimum and adequate potential measurements from inside the tank. The final objective is to be able to map the voltage gradient lines inside the tank bottom from the coupon locations.
12. Number coupons so that coupon # 1 is closest to tank shell while # 3 towards tank center.

Proponent will:

13. Install 3 anodes at optimum spacing from tank and from each other at depths previously decided by CONTRACTOR.
14. De-energize existing CP system. Re-route the anode cables to a portable T/R.
15. Cut 3 (1 m x 1 m) coupons by removing 5 cm wide metallic strips all around the square. Leave the coupons in place completely isolated from the tank (Do not lift the coupons, and do not add water to them nor to the soil gap). The coupons need to be equally spaced along a straight line from the sketch and annular plates towards the tank center.

Trip # 3:

Objective:

Determine:

- 1) coupon corrosion potential.
- 2) coupon-to-coupon potential difference.
- 3) coupon-to-ground contact resistance.
- 4) coupon current pick up.
- 5) 100-mV polarization shift (on coupon).
- 6) voltage gradient mapping outside the tank going away from each anode, radially away from the tank center.
- 7) voltage gradient mapping inside the tank (using cut-out coupon area).
- 8) voltage gradient mapping outside the tank as measured along the circumference between every 2 anodes.

CONTRACTOR will:

16. Physically Ensure complete electric isolation of coupon #1, #2 and #3 (which is closer to tank shell) from the tank bottom (Do not lift the coupon).
17. Add sufficient weight on top of each coupon to ensure improved physical contact with the soil underneath.
18. Try to measure coupon #1, #2 and #3 close free corrosion potential using one (or more) reference electrodes placed inside the (2-in) gap between coupon and tank. Do not pour any water in the gap or anywhere else.
19. Install one data logger per coupon to measure the coupon close and remote potentials.
20. Measure coupon #1, #2 and #3 remote corrosion potential with the reference electrode placed 20-50 m away from the tank.
21. Measure potential difference between coupon 1&2, 2&3, and 3&1.
22. Skip this step if there is no existing CP system. Turn on existing system, and measure the current picked up by the coupons through the digital multimeter set to the ammeter setting. Make sure to connect the leads to the proper terminals of the multimeter. Turn the existing system back to off.
23. While coupons are isolated, measure coupon #1, #2 and #3 contact resistance using the 3-pin Megger method and confirm by measuring the resistance between the coupons and the tank with the 2-pin Megger method
24. Re-confirm resistance measurements by using the 2-pin Megger method between coupons 1 & 2, then 1 & 3, and finally 2 & 3.

25. Energize the temporary CP system and measure remote (where the RE outside the tank) and close on-potentials on the three coupons.
26. Use the digital multimeter in the ammeter mode to measure the current pick up on each of the three coupons
27. Utilize 2 current interrupters to simultaneously: (1) interrupt the CP current from the T/R and (2) isolate coupon from tank. This is to allow obtaining a “true” instant-off polarized potential for the coupon while completely isolated from the tank.
28. Let coupon polarize, take its instant-off potential and repeat the process twice for each coupon.
29. Compare measurements with the free corrosion potential values measured earlier.
30. If more than 100 mV polarization shift between free corrosion and instant-off polarized potential is measured, the coupon is deemed protected. This method is to be used for the first time the test is run; for subsequent or repeat tests use the “depolarization” method.
31. Measure (on) voltage gradient inside tank with 2 RE one remotely placed outside the tank and the other one moving inside the coupon-to-tank gap.
32. Using the voltage shift formulae, back calculate for the effective resistivity as seen by the close anodes placed outside the tank bottom. Conducting this test on a number of tanks and plotting voltage gradient points inside every tank on a graph will generate a trend. This will permit provision for quantifying the effect of the contribution of the resistivity of oily-sand layer to the overall soil resistivity underneath the tank seen by anodes placed outside the tank. This is in addition to the effect on the current distribution and disturbances in the voltage gradient mapping due to the presence of the tank bottom near the anodes.

NOTE: One of the final outcomes of this study would be to develop a voltage gradient behavior for tanks sitting on oily sand pads.

List of Materials Needed for the Test for Each Tank:

- 4 TA-4 anodes
- 16 each 50-lb coke breeze bags
- 200-m # 8 AWG wires
- Portable or existing 50-Amp T/R

Also, Proponent will be expected to:

- Cut three 1 m x 1 m coupons
- Excavate 4 holes for 4 anodes as instructed by CONTRACTOR
- Thermite weld cables and do the field wire connections as instructed by CONTRACTOR

Document History

14 May 2023	Major	Major periodic revision.
01 February 2021	Editorial	Editorial revision to extend due date to 2023.
2 May 2019	Editorial	Editorial revision to correct the reference title of SAES-D-100.
2 April 2018	Major	Reaffirmed the content of the document, and made several changes: • Revised the scope of the standard. • Added the requirement for designer and site personnel qualification and approval. • Elaborated the definitions section in line with other CP standards. • Standard table of Transformer Rectifier Rating added • International reference standards added • Aligned with international standards • SAEP references added • Requirement for A0' sized plot plan drawing showing anode layout, anode gradient circles (anode voltage cones) and UG piping added • Plot plan with hazardous area overlay drawing requirement showing clearly all structures that are CP protected and all CP equipment added • Temperature correction for current density added • Life of permanent CP system increased to 25 years • Soil resistivity survey requirement added • A separate reliable connection either by a bond cable of welded brackets added for valves, piles, etc..., (mechanical joints are not reliable joints for electrical continuity) added. • Installation of isolation devices in hazardous location permitted optionally. • Note on CP equipment in hazardous location added • Included CP coupons in 25% of test stations in plant areas • Use of CP coupons, perforated monitoring tube and stationery reference electrodes for tank external bottom added. • Protection criteria – ON potential upper limit reduced with a commentary note. Criteria for other metals such as CRA added. Limitation on application of polarization shift criteria added. • Note about tank bottom CP for tanks operating at high temperature added. Number of power feed requirements for tank bottoms added.